

Product Announcement

Models: Bosch IDS Light 15 SEER Heat Pump

Brand: Bosch Thermotechnology

Subject: Launch of Bosch IDS Light 15 SEER Heat Pump



BOSCH

Introduction

Bosch Thermotechnology is pleased to announce the launch of the Bosch IDS Light System, the first fully modulating 15 SEER inverter driven condenser available to the market. The model name BOVA15 pairs with the existing BVA15 fixed-speed (PSC) air handlers reaching up to 16 SEER and providing customers with a budget friendly heat pump option from Bosch.

Model, Part Number, and Pricing

There are 3 capacities available for the outdoor unit.

Part Number	Bosch Model	Description	List Price
8733956368	BOVA-24HDN1-M15G	IDS ODU, 24kBTU, Inverter Condensing Unit 15 SEER	\$4,379.00
8733955693	BOVA-36HDN1-M15G	IDS ODU, 36kBTU, Inverter Condensing Unit 15 SEER	\$4,956.00
8733955694	BOVA-60HDN1-M15G	IDS ODU, 60kBTU, Inverter Condensing Unit 15 SEER	\$6,194.00

Features & Benefits

- ▶ Standard efficiency – BOVA15 + BVA15 achieves up to 16 SEER
- ▶ Dual Fuel Ratings Available when paired with Bosch Cased Coil and BGH 96% Furnace
- ▶ Outdoor coil – copper tube with hydrophilic aluminum fins
- ▶ All tonnages use 10 speed ECM outdoor motor for quiet and efficient operation
- ▶ Inverter Drive Compressor (33% - 110% capacity), modulation in 1% increments
- ▶ Whisper Quiet operation – as low as 59 dB
- ▶ Small footprint – 2, 3, & 5 Ton unit offerings
- ▶ Easy to install – compatible with most standard 24 VAC heat pump thermostats
- ▶ Non-communicating – no proprietary thermostat required
- ▶ 10-year Residential Limited Warranty
- ▶ Meets DOE 2023 minimum efficiency requirements

IDS Light Inverter Outdoor Unit



Standard Features

BOVA 15 Features

- ▶ R-410A
- ▶ Inverter Driven Rotary Compressor
- ▶ Compressor Modulation range from 33% - 110%, Capacity in 1% increments
- ▶ Crankcase Heater Standard
- ▶ Multiple System Protection:
 - High pressure switch and low pressure transducer
 - Compressor liquid return protection
 - Compressor high or low compression ratio protection
 - Compressor high temperature protection
 - High / low voltage protection and over current protection
 - IPM and electronic control board high temperature protection
- ▶ Outdoor coil passed 1000 hour salt spray test according to ASTM B117 standard
- ▶ AHRI certified; ETL listed

BOVA 15 Cabinet Features

- ▶ Unique fan-blade design
- ▶ Baked-on powder paint finish
- ▶ Wind load compliant per Florida Building Code - 2010
- ▶ Wire fan discharge grille

Key Components

Key Components of Bosch IDS Light 15 SEER Heat Pump



A Intelligent control features help adjust the output capacity allowing the unit to keep the temperature at your desired level without fluctuation and to keep humidity in check during the summer months. Whether it is summer or winter, the system keeps your home comfortable and the living conditions to your desired taste.

- ▶ The air coil design provides exceptional heat transfer and low air-resistance for high-efficiency operation that can help lower your energy costs.
- ▶ The direct-drive fan blades provide vortex suppression to reduce sound of airflow exiting the condensing section, thereby retaining relatively low noise at high speed - with sound levels as low as 59dBA.
- ▶ The high-efficiency variable capacity inverter compressor can adjust output to any level between 33% and 110%, to perfectly cool and heat your home using minimal energy for maximum comfort.
- ▶ The condensing units come standard with an electronic expansion valve (EEV) to ensure proper refrigerant flow during all conditions in order to optimize the unit's process to the highest efficiency possible.

Individual Unit Content

BOVA 15

- ▶ IDS Light Inverter Outdoor Unit
- ▶ Spare Parts Manual
- ▶ Warranty
- ▶ Installation and Operations Manual

Limited Warranty

For Products installed in a one or two family residential dwelling, BTC warrants that all compressors and internal components incorporated into the Product at the time of shipment by BTC shall remain free from defects in workmanship and materials for 10 years* from the Commencement Date. If the Warranty Registration process has been completed and BTC determines that the Product or any part of the Product has a defect in workmanship or materials, BTC shall pay labor charges associated with the repair or replacement of the part in accordance with the Warranty Labor Allowance Schedule** for the period of 90 days from the Commencement Date.

For Products installed in a building other than a one or two family residential dwelling, BTC warrants that all compressors incorporated into the Product at the time of shipment by BTC shall remain free from defects in workmanship and materials for 3 years* and other internal components incorporated into the Product components for 1 year* from the Commencement Date.

* Please refer to www.bosch-climate.us for full warranty terms and conditions.

** Warranty Labor Allowance Schedule details are available on www.boschprohvac.com

Phase In / Out

- ▶ The launch of BOVA15 will not coincide with any product phase outs.

Certifications

This unit will meet minimum U.S. Federal efficiency requirements and is both AHRI certified and ETL listed.

Literature and Webpage

BOVA15 literature, including the IOM, Unit Warranty, Labor Warranty, Product Specification, Spare Parts List, and Product Brochure will become live on www.boschheatingandcooling.com in the coming weeks.

Technical Data

		BOVA15-24	BOVA15-36	BOVA15-60
Bosch Part Number		8733956368	8733955693	8733955694
Decibels([dB(A)])				
Max @ 100% load		76	78	80
Min @ min load		59	60	60
Compressor				
RLA		19	19	29
LRA		44	44	52
Condenser Fan Motor				
Horsepower (HP)		1/3		
FLA		2.5		
Refrigeration System				
Refrigerant Line Size ¹	Liquid Line Size (OD)	3/8"	3/8"	3/8"
	Suction Line Size (OD)	3/4"	3/4"	7/8"
Refrigerant Connection Size	Liquid Valve Size (OD)	3/8"	3/8"	3/8"
	Suction Valve Size (OD)	3/4"	3/4"	7/8"
	Refrigerant Charge (R410-A, oz)	87	107	152
	Expansion Device	EEV	EEV	EEV
	Maximum Line Length	100 FT	100 FT	100 FT
	Maximum Elevation Difference	50 FT	50 FT	50 FT
Operating Range				
Cooling		40°F-120°F		
Heating		5°F-86°F		
Electrical Data				
Voltage-Phase-Hz		208/230-1-60	208/230-1-60	208/230-1-60
Minimum Circuit Ampacity ²		22.9	26.7	39.2
Max. Overcurrent Protection ³		35	45	60
Min/Max Volts		172V/270V		
Weight				
Net Weight (without packaging) lbs		126	134	186
Gross Weight (including packaging) lbs ⁴		154	162	218
Dimensions				
Unit L x W x H (in.)		28 x 28x 24-15/16	28 x 28x 24-15/16	29-1/8 x 29-1/8 x 33-3/16
Outdoor Coil				
Net face area - sq.ft. Outer Coil		12.7	12.7	18.3
Tube diameter-in.		9/32" (7mm)	9/32" (7mm)	9/32" (7mm)
No.of rows		1	2	2
Fins per inch		19	17	19

¹ Tested and rated in accordance with AHRI Standard 210/240.

² Wire size should be determined in accordance with National Electrical Codes; extensive wire runs will require larger wire sizes.

³ Must use time-delay fuses or HACR-type circuit breakers of the same size as noted.

⁴ Weight values are estimated.

AHRI 210/240 Performance Data

Inverter Ducted Split AHRI 210/240 Performance Data										
System Configuration	Outdoor Unit Model	Indoor Unit Model	Furnace Model	Cooling Capacity (BTU/h)			Heating Capacity			CFM
		Coils/Air Handlers		Total	EER ¹	SEER ²	Hi	HSPF ³	Low ⁴	
BOVA 15 with BVA15	BOVA-24HDN1-M15G	BVA-24WN1-M15	/	24000	11.2	15.5	24000	9.0	20000	830
	BOVA-36HDN1-M15G	BVA-24WN1-M15	/	24000	11.8	16.0	24000	9.0	20600	830
	BOVA-36HDN1-M15G	BVA-36WN1-M15	/	34200	10.4	15.0	35000	9.0	25000	1220
	BOVA-60HDN1-M15G	BVA-48WN1-M15	/	46500	11.2	15.0	48000	9.0	38000	1530
	BOVA-60HDN1-M15G	BVA-60WN1-M15	/	50000	10.2	15.0	56000	9.0	40500	1810
	BOVA-24HDN1-M15G	BMAC2430ANTD	BGH96M060B3B	23600	11.5	16.5	24000	9.0	17600	720/630
BOVA 15 with 96% Gas Furnace	BOVA-24HDN1-M15G	BMAC2430ANTD	BGH96M080B3B	23600	11.5	16.5	24000	9.0	17600	700/610
	BOVA-24HDN1-M15G	BMAC2430BNTD	BGH96M060B3B	24000	11.5	16.5	24000	9.0	17600	720/630
	BOVA-24HDN1-M15G	BMAC2430BNTD	BGH96M080B3B	24000	11.5	16.5	24000	9.0	17600	700/610
	BOVA-36HDN1-M15G	BMAC3036ANTD	BGH96M060B3B	33000	10.0	16.5	34200	8.8	23000	1000/750
	BOVA-36HDN1-M15G	BMAC3036ANTD	BGH96M080B3B	33000	10.0	16.5	34200	8.8	23000	1000/750
	BOVA-36HDN1-M15G	BMAC3036BNTD	BGH96M060B3B	33600	10.0	16.5	34200	8.8	23000	1050/770
	BOVA-36HDN1-M15G	BMAC3036BNTD	BGH96M080B3B	33600	10.0	16.5	34200	8.8	23000	1050/770
	BOVA-36HDN1-M15G	BMAC3036CNTD	BGH96M080C4B	33600	10.0	16.5	34200	8.8	23000	1030/750
	BOVA-36HDN1-M15G	BMAC3036CNTD	BGH96M100C5B	34000	10.0	16.5	34200	8.8	23000	1150/750
	BOVA-60HDN1-M15G	BMAC4248BNTF	BGH96M080B3B	43000	10.0	16.5	45000	8.8	31000	1250/1050
	BOVA-60HDN1-M15G	BMAC4248CNTF	BGH96M080C4B	43000	10.5	16.5	45000	8.8	31000	1470/1200
	BOVA-60HDN1-M15G	BMAC4248CNTF	BGH96M100C5B	45000	10.5	16.5	46500	8.8	32000	1450/1150
	BOVA-60HDN1-M15G	BMAC4248DNTF	BGH96M100D5B	45500	10.5	16.5	47000	8.8	32000	1500/1200
	BOVA-60HDN1-M15G	BMAC4248DNTF	BGH96M120D5B	45500	10.5	16.5	47000	8.8	32000	1500/1200
	BOVA-60HDN1-M15G	BMAC4860CNTF	BGH96M100C5B	53000	9.5	16.0	53500	9.0	38000	1660/1150
	BOVA-60HDN1-M15G	BMAC4860DNTF	BGH96M100D5B	53500	10.0	16.5	54000	9.0	38000	1700/1200
	BOVA-60HDN1-M15G	BMAC4860DNTF	BGH96M120D5B	53500	10.0	16.5	54000	9.0	38000	1700/1200

¹ Energy Efficiency Ratio; Certified per AHRI 210/240

² Seasonal Energy Efficiency Ratio; Certified per AHRI 210/240³

³ HSPF = Heating Seasonal Performance Factor; Certified per AHRI 210/240

⁴ Jumper cut or dip switch off

AHRI 210/240 Performance Data (SEER2/EER2/HSPF2) - Tested to New DOE 2023 Standard

Inverter Ducted Split AHRI 210/240 Performance Data										
System Configuration	Outdoor Unit Model	Indoor Unit Model	Furnace Model	Cooling Capacity (BTU/h)			Heating Capacity			CFM
		Coils/Air Handlers		Total	EER2 ¹	SEER2 ²	Hi	HSPF2 ³	Low ⁴	
BOVA 15 with BVA15	BOVA-24HDN1-M15G	BVA-24WN1-M15	/	24000	10.6	14.5	24000	8.0	20000	780
	BOVA-36HDN1-M15G	BVA-24WN1-M15	/	24000	11.2	14.8	24000	8.2	20600	780
	BOVA-36HDN1-M15G	BVA-36WN1-M15	/	34200	10.0	14.5	34600	8.0	25000	1210
	BOVA-60HDN1-M15G	BVA-48WN1-M15	/	46500	10.6	14.3	48000	8.0	38000	1530
	BOVA-60HDN1-M15G	BVA-60WN1-M15	/	54000	9.8	14.3	56000	8.0	40500	1680
	BOVA-24HDN1-M15G	BMAC2430ANTD	BGH96M060B3B	23600	10.5	16.0	24000	8.0	17600	750/560
BOVA 15 with 96% Gas Furnace	BOVA-24HDN1-M15G	BMAC2430ANTD	BGH96M080B3B	23600	10.5	16.0	24000	8.0	17600	750/560
	BOVA-24HDN1-M15G	BMAC2430BNTD	BGH96M060B3B	24000	10.5	16.0	24000	8.0	17600	750/560
	BOVA-24HDN1-M15G	BMAC2430BNTD	BGH96M080B3B	24000	10.5	16.0	24000	8.0	17600	750/560
	BOVA-36HDN1-M15G	BMAC3036ANTD	BGH96M060B3B	33000	9.0	15.5	34200	7.8	23000	1050/750
	BOVA-36HDN1-M15G	BMAC3036ANTD	BGH96M080B3B	33000	9.0	15.5	34200	7.8	23000	1050/750
	BOVA-36HDN1-M15G	BMAC3036BNTD	BGH96M060B3B	33600	9.0	15.5	34200	7.8	23000	1050/780
	BOVA-36HDN1-M15G	BMAC3036BNTD	BGH96M080B3B	33600	9.0	15.5	34200	7.8	23000	1050/780
	BOVA-36HDN1-M15G	BMAC3036CNTD	BGH96M080C4B	33600	9.0	15.5	34200	7.8	23000	1050/780
	BOVA-36HDN1-M15G	BMAC3036CNTD	BGH96M100C5B	33600	9.0	15.5	34200	7.8	23000	1050/650
	BOVA-60HDN1-M15G	BMAC4248BNTF	BGH96M080B3B	43000	9.0	16.0	44000	7.8	30000	1100/950
	BOVA-60HDN1-M15G	BMAC4248CNTF	BGH96M080C4B	43000	9.5	16.0	44000	7.8	30000	1150/950
	BOVA-60HDN1-M15G	BMAC4248CNTF	BGH96M100C5B	45000	9.5	16.0	47000	7.8	32000	1600/1100
	BOVA-60HDN1-M15G	BMAC4248DNTF	BGH96M100D5B	45500	9.5	16.0	47000	7.8	32000	1600/1100
	BOVA-60HDN1-M15G	BMAC4248DNTF	BGH96M120D5B	45500	9.5	16.0	47000	7.8	32000	1600/1100
	BOVA-60HDN1-M15G	BMAC4860CNTF	BGH96M100C5B	53000	9.0	15.5	53500	8.0	38000	1700/1350
	BOVA-60HDN1-M15G	BMAC4860DNTF	BGH96M100D5B	53500	9.5	15.5	54000	8.0	38000	1700/1400
	BOVA-60HDN1-M15G	BMAC4860DNTF	BGH96M120D5B	53500	9.5	15.5	54000	8.0	38000	1700/1400

¹ Energy Efficiency Ratio; Certified per AHRI 210/240

² Seasonal Energy Efficiency Ratio; Certified per AHRI 210/240³

³ HSPF = Heating Seasonal Performance Factor; Certified per AHRI 210/240

⁴ Jumper cut or dip switch off

Frequently Asked Questions

Q1: Will BOVA15 be DOE 2023 compliant?

A1: Yes, the BOVA15 will be released with both SEER1 and SEER2 ratings and will meet the minimum efficiency requirements effective Jan 1, 2023.

Q2: What are the line set and length and limitations?

A2: Line set cannot exceed 100' length nor a lift of 50' (if 50' up then needs to be 50' out). Refer to IOM based on lineset length to determine max lift.

Q3: Do BCC50 and BCC100 work with IDS Light?

A3: Yes, both thermostats can be used with IDS Light. See wiring diagrams for more details.

Q4: Is a Bosch thermostat necessary to install with the Bosch IDS (Inverter Driven System)?

A4: No, Bosch IDS family of equipment does not require proprietary controls, as the logic of the equipment is designed into the outdoor condensing section.

Q5: What is the main difference between IDS Light and other inverter products like it on the market?

A5: Independent compressor modulation maintains constant indoor coil temperature, allowing for excellent dehumidification without the need for indoor fan modulation.

Q6: What are the major differences in design between the IDS Premium and Plus Systems vs. IDS Light?

A6: The IDS Light outdoor unit has 3 available SKUs, 2ton, 3ton, and 5ton and matches with the BVA15 to create a budget friendly inverter driven system with all the same benefits of comfort that is realized by inverter technology. The 2 and 3 ton units share the same cabinet size, while the 5 ton is slightly larger.

Q7: Is there a new air handler being launched with IDS Light outdoor unit?

A7: No, the IDS light can pair with the existing BVA15.